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Multi-Featured Movie Recommendation Using Knowledge Graph

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***Abstract*—Recommendation systems are crucial in offering items, particularly in streaming services. Recommendation algorithms are essential for streaming movie services like Netflix in aiding consumers in identifying new movies to watch. Provide a Technologically advanced strategy on auto-encoders for constructing a movie filter system that anticipates users' ratings for movies through vast end-users rating data. Examine the application of deep learning to predict customers' assessments of new movies using the Movie-Lens dataset, providing movie recommendations. The movie recommendation system proposed here uses particle filtering embedded in a knowledge graph. The model enables the user to provide the best suitable recommendation of movies considering the user's favorite directors and genre. The experimental findings reveal that compared by error, our recommendation system beats a user-based model on projected ratings and even in the survey where people assess between concepts from both systems.**

***Index Terms*—Movie Recommendation, Knowledge Graph; Machine Learning; Particle Filtering**

I. Introduction

Recommendation systems are used to predict the user's rating or preference for an item. Recommender Systems vary from collaborative filtering methods to the currently trending neural network-based methods [1], [2]. Different approaches incorporate KGs in recommendation systems. Mainly the work focus ranges from external KGs to supplement content-based recommendations with those entirely powered by KGs [3].

A recommendation system is widely used in the entertainment domain to suggest videos, music, and movies [4]. Among them, the preferred mode of entertainment is movies [5]. The online platforms use sound recommendation systems to recommend movies to their users. Movie Recommendation Systems allow us to look for our favored films amongst all those distinct varieties of films and consequently lessen the hassle of spending plenty of time looking for a movie according to our preference. It is a subclass of information filtering to predict the preferences of movies for users.

Although there are many approaches developed in the past, search still goes on due to it being often used in many applications, which personalize recommendations and deal with information overload. Many recommendation systems have been developed using collaborative, content-based, or hybrid filtering methods. The proposed system uses content-based filtering using Knowledge Graphs and Machine Learning concepts.

Recommendation systems are becoming more and more common in our lives. The goal of Movie recommendation is to make user-friendly and preferred recommendations of movies [6]. The genre and other movie tags provide a basis for content-based filtering [7]. Cosine similarity classification produces the most accurate results; this approach combines different genres, and tags [8]. Movies are compared with a high rate of accuracy, which helps to determine the user's interest more precisely. Cosine similarity is used to judge the similarity measure between the two movies [9], [10].

Different recommender system has are proposed over the last decade [11]. A few KG-based systems are evolved. We focus on particle filtering based on eigenvector centrality to generate user recommendations [12]. The approach appears efficient and straightforward [13] [14] to make user-friendly and preferred recommendations of movies [6]. The genre and other movie tags provide a basis for content-based filtering [7]. Cosine similarity classification produces the most accurate results; this approach combines different genres, and tags [8]. Movies are compared with a high rate of accuracy, which helps to determine the user's interest more precisely. Cosine similarity is mostly used to judge the similarity measure between the two movies [9], [10]. A few KG-based systems are evolved. We focus on particle filtering based on eigenvector centrality to generate user recommendations [12]. The approach appears efficient and straightforward [13].

II. Literature Review

Knowledge graph-based recommendation system is powerful than memory-based recommendation systems. Graph-based data models enable relationship-based queries in real time. Graph-based data models enable understanding behavioral activities like customers' past purchases and likes or dislikes. Knowledge graphs are used in product/ service recommendations, question answering [14], know-your-customer (KYC), anti-money laundering initiatives, and better decisions using data from the ground.

The authors in [15] studied numerous recommendation systems and general information. The authors examined both personalized and non-personalized recommendation

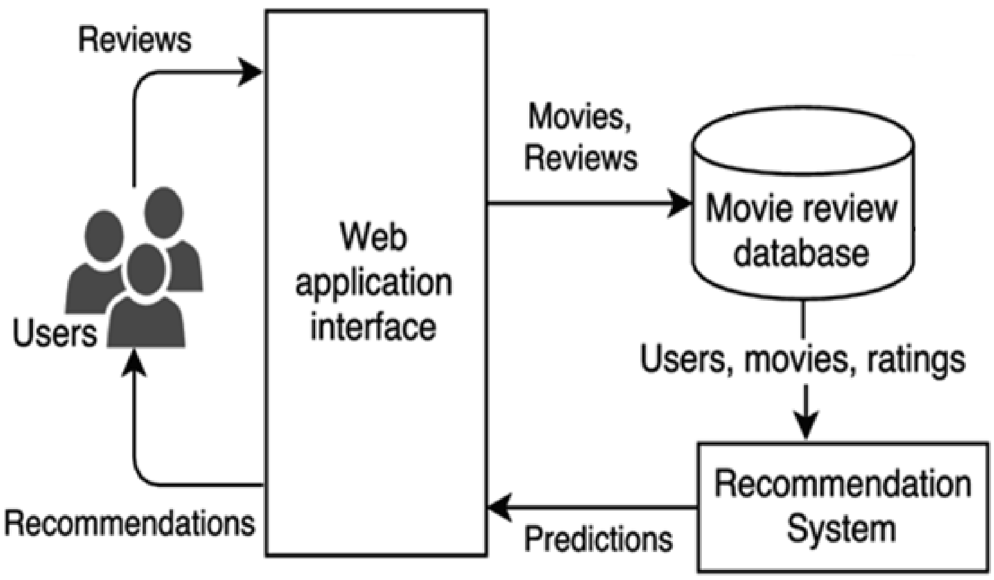

Fig. 1. Architecture of Movie Recommendation System

systems. A great example was provided to explain user-based vs. item-based collaborative filtering. The authors also evaluated the benefits and detriments of different recommendation systems. The authors in [16] evaluated Simple Prediction Models, Content-based Recommender Systems, and Collaborative Filtering-based Recommender Systems before recommending a Hybrid Recommendation System as a solution. The authors considered cosine similarity and Singular Value Decomposition(SVD), using which the algorithm provides 30 movie selections. They subsequently filter these videos based on SVD and user ratings.

The article, [17] suggested a K-means clustering algorithm approach. The authors employed clusters to separate similar users. The authors developed a neural network for each cluster for recommendation applications. Data Pre-processing, Principal Component Analysis, Clustering, Data Analysis for Neural Networks, and Designing Neural Network are all processes in the proposed system. User rating, user preference, and consuming user ratio were all considered. After the grouping stage, the scientists deployed a neural network to predict the ratings the user may give the un-watched movies. Finally, recommendations are made using projected high ratings.

The authors in [18] have developed a collaborative filtering strategy for personalized movie choices. The Euclidean distance metric was used to select the most similar user. The user with the least Euclidean distance is identified. Finally, movie assumptions are based on what the user rated the highest. The authors even claim that the recommendations alter over time so that the program can keep up with the changing likes of the user. Despite substantial research on recommendation systems, development is still needed to solve present problems. The authors in [19] developed a hybrid recommendation technique that hierarchically employs a content-based filtering technique and an information retrieval approach to give customers personalized movie suggestions. The authors have developed movie proposals utilizing a perfect sequence of photographs that describe the movie narrative plot, which is a unique part of this research work. This enhances the visuals. The recommended technique is separated into four major sections.

Initially, networking sites such as Facebook are used to learn about users' interests. The movie evaluations must be examined, and recommendations must be given. Finally, the narrative plan must be produced to have superior visuals. Consequently, hunting for and accessing facts to make judgments and reach validated measures is challenging. Consequently, the user must filter through many results to discover the one that he or she requires. This may be utilized when browsing for books, music, films, and job adverts. A recommendation system is consequently necessary to aid in proposing things to users that are more relevant and accurate, as well as satisfy the user's wishes and expectations. A movie recommendation system proposes movies comparable to the user's likes and preferences.

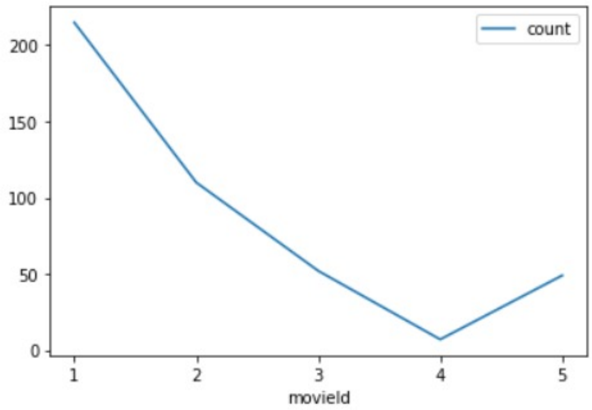

Fig. 2. Movie Rating for recommendation

III. The Proposed System

A. Movie Recommendation

A complex machine-learning model provides suggestions to the users. Users are given recommendations based on a derivative of the eigenvector algorithm to utilize its maximum potential in this study. Our system ranks the most valuable and interesting web pages by analyzing their connections. This paper optimizes an approximate algorithm to compute the highly influential Top-K graph nodes.

Recommendation systems have become ubiquitous in recent years as they deal with the information overburden problem by suggesting users the most significant products from an immense amount of data [1]. Movie recommendation systems aim to help movie admirers by suggesting what movie to watch without having to go through the long process of choosing from a huge set of movies that go up to thousands and millions, which is time-consuming and disconcerting. Our ultimate intention is to scale down the human endeavor by suggesting movies based on the user's enthusiasm. The system architecture of the proposed system is shown in Fig. 1.

Recommendation systems are one of the most successful and boundless applications of machine learning. With this, the conferred recommendation system generate proposal using various types of knowledge and data about users from the movie dataset. The user can flip through the recommendations easily and find a movie. The recommendation systems are decisive as they help them make the right choices without depleting their cognitive capital. A recommendation system aims to look for content that is frantic to an individual. Moreover, it includes several things to create customized lists of valuable and interesting content exclusive to every user/individual.

B. Methodology

With the proliferation of information, the ability to quickly find a favorite movie among many options has become a critical issue. A customized recommendation system can be beneficial when a consumer does not have a specific movie in mind. The methods implemented in this paper are knowledge graphs and content-based filtering.

1) Content-Based Filtering: Content-Based Filtering deals with the delivery of items selected from an extensive collection that the user is likely to find interesting or valuable and is a classification task which in this case are movies with similarities [20]. The basic process is performed by matching user preferences with item attributes. It selects items based on the interconnection between the chosen items' content and users' preferences. During the construction of the Movie recommendation system, the genre of the movie, the actors present in it, or even the movie's director are considered. This system uses one or more combinations of textures. In this, the system has been built on the form of genres that the user might opt to watch. So content-based filtering is the approach adopted to do this. Content-based recommendation systems per pends the user's precedent nature and identifies patterns in them to propose similar items.

2) Knowledge Graph: A knowledge graph is a perception layer of linked data enriched with semantics, so you can reason with the underlying data and use it for complex decision-making. We bridge diverse and disparate data sets regardless of data type, such as structured, unstructured, and semi-structured. The data is then mapped, which draws connections among them for the first layer of dynamic context, which provides immediate understanding. We are to apply semantics than give deeper context to corresponding data. The deeper the context, the more meaningful the insights are.

Knowledge graph can effectively standardize and exhibit knowledge so that is exercised in advanced applications. A key challenge in producing knowledge graphs is rationally collecting noisy information from different sources.

C. Date Pre-Processing

Data being a collection of objects and attributes, data is pre-processed before making a fully functional system. The data frequently needs to be pre-processed to be used by the machine learning techniques in the dissection stage. The movie data set collected redundant and null data, which might be left out points, transcribed data, or clatter. Data cleaning and transformation are used to eliminate outliers and provide the data with a form that makes it simple to build models. After standardization and cleaning techniques, the dataset is fed into the system to get a recommendation model. The final stage of data pre-processing in machine learning is feature scaling. It is a method for standardizing the dataset independent variables within a given range. When feature scaling, we place our variables in the same field so that no one variable dominates the others. A feature extraction process is implemented on the dataset.

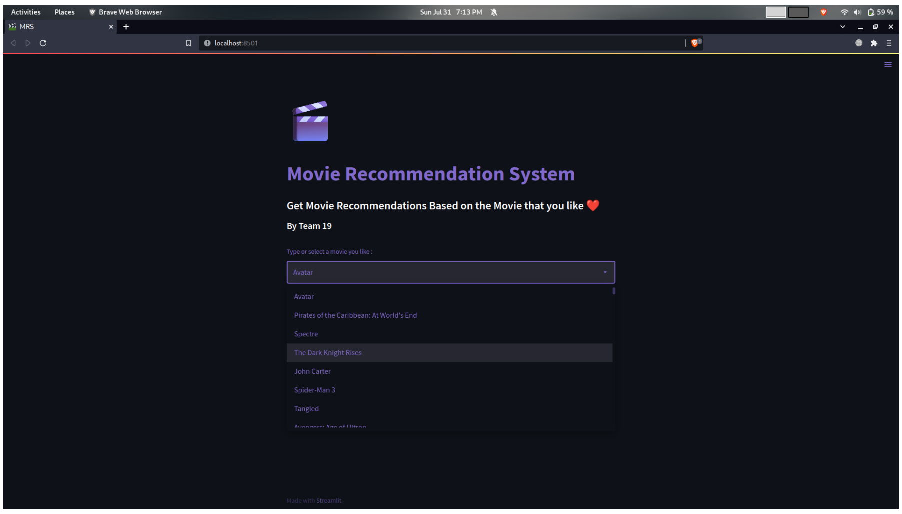

Fig. 3. Selecting a movie.

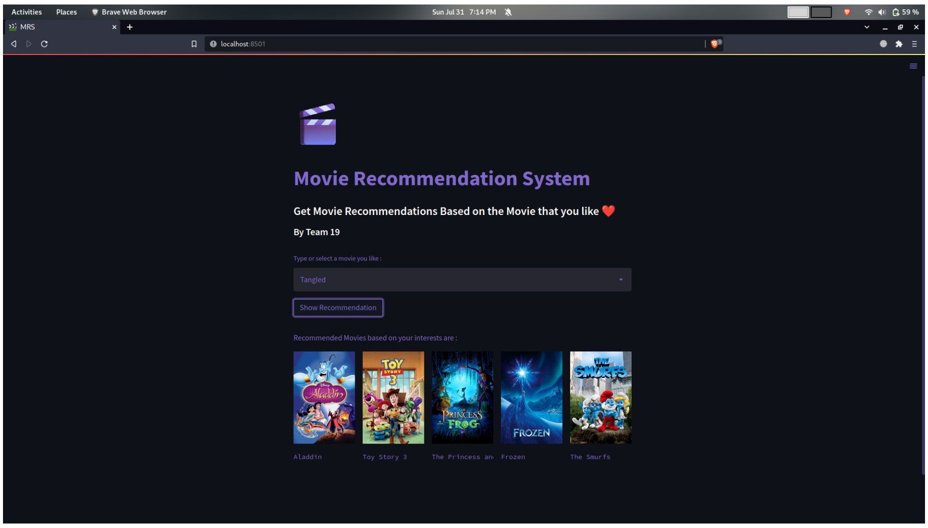

Fig. 4. Five Similar Movies being Displayed.

IV. Experimentation

We aim to categorize the movies according to different criteria and propose a film(s) from the dataset by applying deep learning algorithms. It is a web-mining technique where similarity metrics are used for recommendation. The algorithm works based on co-related items and their variance. The best example is the Pearson correlation.The degree to which the similarity evaluated on a scale of -1 to +1.

A. Experiment

Handling five thousand set of data comes with its own set of challenges, so first the data was subjected to a basic test where the relations were checked and then packaged into a pickle file. Later tests were done in reading this file through streamlit where a movie was searched for and the result were thus obtained. Each of the output were verified to confirm whether they were similar to the one searched for. Care was also given in checking if all the displayed contents and input boxes were properly aligned and displayed.

B. Dataset

According to its description, the MovieLens (Small) dataset contains a hundred thousand ratings, over three thousand tags, and around nine thousand flicks. With a graph database, we have fast access to both data (user, movie, genre) and relationships between them, which allow us to process queries very fast, enabling using the model for real-time recommendation engines. Several works of the literature suggest novel similarity metrics that include the Jaccard index in existing measures. These techniques undoubtedly compensate for the shortcomings of prior similarity metrics, primarily due to data sparsity.

Another benefit of utilizing a graph database for this model is that it is easier to view and comprehend the connections and routes that lead to suggestions.

V. Result and Discussion

Our recommendation model suggests movies based on their earlier ratings. Many data, such as ratings and genres, are used to recommend new movies to users. Because of information saturation, recommendation systems system has grown increasingly crucial. The proposed approach help to increase the accuracy of the recommender system, the movie in particular for the content-based recommendation systems. This graph, Fig. 2 depicts the count of ratings of movies. The extensive collection of recommendations and forecasts is stored in the database. If the forecast is incorrect, no recommendation is exercised. The graph depicts the relationship between movies and the estimates are given by the recommendation algorithm.

A recommendation system solves this problem by allowing users to acquire knowledge, products, and services on a personalized basis. The Recommendation System is built using a dataset containing a movie database of five thousand movies and utilizing content-based filtering to get a list of similar movie recommendations per the user's selection.

VI. Conclusion and Future scope

In this fast-paced and chaotic world where it is difficult for you to rejoice, entertainment has always been the escape. Movie Recommendation System thus provides a fast and easy alternative to lessen the hassle and find a movie that a person might prefer. The illustrated modeling of a movie recommendation system was made by making use of the concept of content-based filtering by considering a data set that includes a collection of 5000 Movies from TMDB and performing detailed data analysis as well as data processing. Content-based filtering selects movies based on the correlation between the contents and tags of the film. A knowledge graph was built to ensure that the model is accurate and further helps visualize the flow of information in addition to summarising relationships. The model can be developed to a mobile application in future so that new users can select movies and their ratings for movies and later rate the recommendation at finger tip.

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